

Machine-Learning-Based Link Adaptation Across Heterogeneous 5G NR Channel Configurations

Rohan Marwaha
Under the supervision of Dr. Zhao

Abstract—This paper investigates machine-learning-based link adaptation for predicting the best modulation and coding scheme (MCS) across heterogeneous 5G New Radio channel configurations. A consistent evaluation pipeline is developed using Channel Quality Indicator (CQI), Rank Indicator (RI), Precoding Matrix Indicator (PMI), and available channel descriptors. Multiple classifiers are first benchmarked on representative FR1 and FR2 datasets, including decision trees, linear support vector machines, random forests, extremely randomized trees, a multilayer perceptron, and XGBoost. In the BankB feature comparison, the strongest preliminary result is obtained using tuned XGBoost with CQI-only input. On the BankB FR2 wideband dataset, CQI-only XGBoost achieves 65.93% exact accuracy and 90.63% accuracy within ± 1 MCS. A cross-bank evaluation from BankA to BankG shows exact accuracy ranging from 57.60% to 75.88% and within- ± 1 accuracy ranging from 87.65% to 92.60%. The results are consistent with the hypothesis that CQI captures much of the useful information derived from channel state information. Differences across banks are associated with changes in system configuration, propagation environment, signal-to-noise ratio range, Doppler conditions, and target-label distribution, although controlled experiments are still needed to isolate individual effects.

Index Terms—5G NR, link adaptation, modulation and coding scheme, CQI, XGBoost, machine learning, FR1, FR2, mmWave.

I. INTRODUCTION

Link adaptation selects a modulation and coding scheme according to current channel conditions. The objective is to maximize throughput while maintaining an acceptable block error rate. Traditional link adaptation relies on channel-quality reports and threshold-based mappings, but the relationship between measured channel indicators and the best MCS can become complex across different propagation environments, bandwidths, carrier frequencies, Doppler conditions, and antenna configurations. Recent work has therefore explored supervised learning, reinforcement learning, and online deep learning for MCS selection and link adaptation [1]–[5].

This work evaluates supervised machine-learning models for best-MCS classification using a multi-bank 5G NR dataset. The study begins with model benchmarking on BankA and BankB, then compares CQI-only, CQI/RI/PMI, and full feature sets. Finally, a tuned CQI-only XGBoost pipeline is evaluated across BankA through BankG to analyze the effect of system configuration on prediction performance.

The main contributions are as follows:

- A consistent train, validation, and test pipeline for best-MCS classification across multiple NR configuration banks.
- A comparison of tree-based models, linear SVM, a neural network, and tuned XGBoost using exact, within- ± 1 , over-prediction, and under-prediction metrics.
- A feature-set analysis showing that CQI-only input is sufficient to match or exceed the performance of CQI/RI/PMI and full channel-descriptor inputs on BankB.
- A cross-bank study showing how prediction accuracy varies across FR1, FR2, indoor mmWave, outdoor line-of-sight, S-band, L-band, and fixed-wireless-access configurations.

II. BACKGROUND AND RELATED WORK

A. CSI-Derived Link Adaptation Indicators

Channel State Information (CSI) describes the estimated propagation channel used by a transmitter or receiver to support link adaptation. In 5G NR, channel-quality and precoding feedback is reported through quantities including CQI, RI, and PMI, and the physical-layer procedures and MCS tables are specified by 3GPP TS 38.214 [10]. RI summarizes the estimated spatial rank, while PMI identifies a preferred precoding matrix. CQI is a compact channel-quality report associated with a supportable transmission format. Because these indicators are derived from the same underlying channel estimate, they can contain overlapping information for a best-MCS classifier.

B. Supervised Learning for MCS Prediction

Direct supervised MCS prediction is the closest prior-work category to the present study. Tsipi *et al.* compared several machine-learning classifiers, including neural networks, support vector machines, random forests, and bagged nearest-neighbor models, for multiclass MCS prediction in a simulated 5G setting [1]. Their study demonstrates the feasibility of replacing fixed mappings with data-driven classifiers and provides a useful reference for classifier and feature comparisons. Pati *et al.* also proposed a machine-learning approach to link adaptation in 5G NR, supporting the broader use of learned decision rules for selecting transmission settings [2]. In contrast, the present work emphasizes a reusable CQI-only XGBoost pipeline and compares performance across heterogeneous configuration banks rather than optimizing one fixed scenario.

Martín-Vega *et al.* reviewed emerging 5G NR link-adaptation tools and described link adaptation as a multi-domain problem affected by numerology, bandwidth part selection, channel estimation, carrier frequency, propagation conditions, and scheduling constraints [3]. This perspective

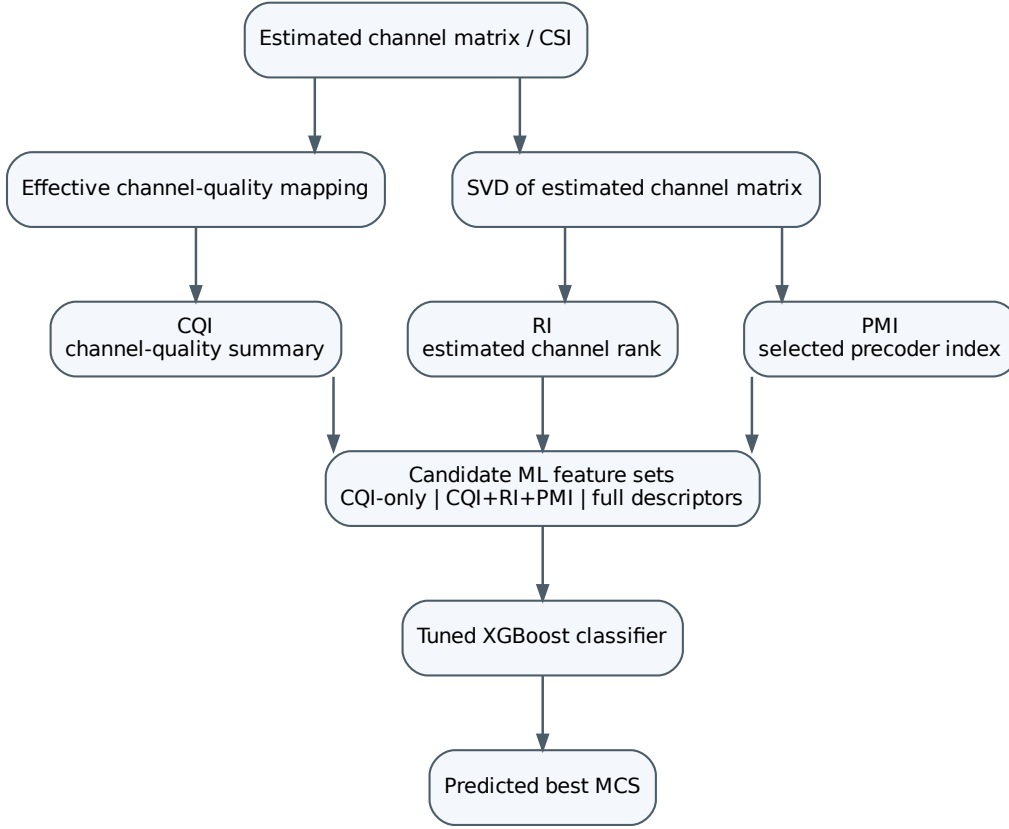


Fig. 1. Conceptual relationship between CSI-derived indicators, the evaluated feature sets, and best-MCS prediction. The diagram is an explanatory abstraction rather than a complete 3GPP signal-processing implementation.

motivates the cross-bank analysis in this paper, where configuration changes are evaluated using a consistent prediction and metric pipeline.

C. Reinforcement, Online, and Feedback-Aware Adaptation

Other work formulates link adaptation as a sequential or continuously updated decision problem. Mota *et al.* used Q-learning to learn a CQI-to-MCS policy that balances spectral efficiency against block error rate [4]. Bobrov *et al.* developed an online deep-learning method for massive-MIMO adaptive modulation and coding, initializing a neural model from a conventional algorithm and updating it using service feedback [5]. These methods can adapt during operation, while the present study uses offline supervised multiclass classification to isolate how feature sets and system configurations affect prediction accuracy.

CQI can also become stale when feedback is delayed. Balieiro *et al.* estimated updated channel quality from user context, SINR, and delay information using machine-learning and evolutionary-computing approaches [7]. Their result is particularly relevant to the planned satellite extension, where

faster channel variation and high Doppler may reduce the usefulness of delayed CQI even if CQI is highly informative in the current terrestrial data banks.

D. Error Asymmetry and Empirical Wireless Data

Rokoni *et al.* evaluated machine-learning-based MCS adaptation using empirical sidelink measurements and compared random forests, gradient boosting, and neural approaches [6]. Their discussion highlights an operational asymmetry: selecting an MCS above the channel capability can make a packet undecodable, whereas a conservative MCS generally sacrifices data rate. This distinction supports reporting separate over-prediction and under-prediction rates in addition to exact accuracy. The present paper further adds a within- ± 1 MCS metric to distinguish near misses from larger classification errors.

E. Non-Terrestrial Network Context

Azari *et al.* surveyed the evolution of non-terrestrial networks from 5G to 6G and discussed satellite mobility, Doppler, propagation delay, time-varying channels, and integration with

terrestrial systems [8]. These challenges motivate the planned BankH–BankK extension, which will compare S-band and Ka-band satellite configurations under clear-sky and rain-fade conditions. The satellite analysis is not yet part of the current numerical results and will require newly generated datasets.

Within this selected literature set, the studies establish the value of supervised, reinforcement, and online learning for link adaptation, but none performs the same CQI-only tuned-XGBoost comparison across the project-specific FR1, FR2, indoor and outdoor mmWave, S-band, L-band, and FWA banks while jointly reporting exact accuracy, within- ± 1 accuracy, over-prediction, under-prediction, and gain over a bank-specific majority baseline. This is a gap within the reviewed papers, not yet a universal novelty claim; a broader systematic search is required before making a stronger statement.

III. DATASET AND SYSTEM CONFIGURATIONS

The dataset contains 20,000 samples and 121 columns across multiple system banks. Each bank contains bank-specific input features and a target column named `bestMCS_{bank}`. The observed MCS values lie between 0 and 27. The number and identity of observed classes vary by bank, with up to 23 classes in the evaluated configurations. The multi-bank dataset was generated by the project system team and provided for this study. Individual dataset-generation contributors should be named in the final acknowledgment after their full names and preferred attribution are confirmed.

IV. METHODOLOGY

A. Feature Sets

Three feature sets are evaluated:

- 1) CQI-only.
- 2) CQI, RI, and PMI.
- 3) Full features: CQI, RI, PMI, estimated RMS delay spread, frequency selectivity, and coherence time when available.

B. Models

The evaluated models are a most-frequent-label baseline, decision tree, linear SVM, random forest, ExtraTrees, a PyTorch multilayer perceptron, and XGBoost. XGBoost uses the native training API, label remapping to contiguous class indices, a validation split, and early stopping. Three parameter configurations are compared, including shallow, regularized, and conservative variants. The implementation follows the gradient-boosted tree framework introduced by Chen and Guestrin [9].

C. Evaluation Protocol

An outer 80/20 stratified train/test split is used. For XGBoost and the neural network, 20% of the outer training partition is used for validation, corresponding to 64% training, 16% validation, and 20% testing relative to the full dataset. Missing, positive-infinite, and negative-infinite feature values are replaced with zero before splitting. Standardization is applied only to the linear SVM and neural network. Class

weighting is used for the decision tree, linear SVM, random forest, ExtraTrees, and neural network, while the current XGBoost implementation is unweighted. All random seeds are fixed for repeatability.

For publication-quality results, XGBoost configurations must be selected using validation data only and evaluated once on the untouched test set. The numerical results reported in this draft were generated by the initial pipeline and should be treated as preliminary until they are reproduced with the corrected validation-only tuning procedure and, preferably, multiple random seeds.

For N test samples with true MCS y_i and predicted MCS $\hat{y}_i$, exact accuracy is

$$A_{\text{exact}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\hat{y}_i = y_i). \quad (1)$$

Accuracy within ± 1 MCS is

$$A_{\pm 1} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(|\hat{y}_i - y_i| \leq 1). \quad (2)$$

The over-prediction and under-prediction rates are

$$R_{\text{over}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\hat{y}_i > y_i), \quad (3)$$

$$R_{\text{under}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\hat{y}_i < y_i). \quad (4)$$

D. Analysis Flow

Figure 2 summarizes the validated evaluation procedure. Feature preprocessing and label mapping are completed before a stratified split. Hyperparameter selection and early stopping use validation data only, after which the selected configuration is retrained on the full outer training partition and evaluated once on the untouched test set.

V. RESULTS

A. BankA Full-Feature Model Benchmark

Using the full feature set on BankA_FR1_sub6, the most-frequent-label baseline achieves approximately 20.4% exact accuracy. The decision tree reaches approximately 56.3% exact accuracy and 86.2% within ± 1 . The linear SVM reaches approximately 46.6% exact accuracy, while the basic MLP reaches approximately 52.6% exact accuracy and 90.4% within ± 1 . Initial XGBoost performance is approximately 51–52% exact accuracy, but tuning with early stopping increases performance to approximately 56.5% exact accuracy and 85.7% within ± 1 .

B. BankB Model Benchmark

Tuned XGBoost provides the highest exact accuracy, while the DNN provides the highest within- ± 1 accuracy among the full-feature models. The baseline is relatively high because MCS 27 is the dominant class.

TABLE I
SELECTED LITERATURE CLOSELY RELATED TO THE PRESENT STUDY

Reference	Scenario and method	Inputs and target	Main relevance	Difference from this work
Tsipi <i>et al.</i> [1]	Simulated 5G; ANN, SVM, RF, and bagged k-NN	Physical-layer measurements; multi-class MCS	Direct benchmark for supervised MCS prediction and feature selection	No CQI-only tuned-XGBoost comparison across the present heterogeneous banks
Pati <i>et al.</i> [2]	5G NR machine-learning link adaptation	Channel measurements; link-adaptation decision	Establishes data-driven 5G NR link adaptation	Short study without the present cross-bank metric analysis
Martín-Vega <i>et al.</i> [3]	Survey and analysis of 5G NR link-adaptation tools	Multiple physical and system domains	Motivates configuration-aware analysis	Broad framework rather than a single reproducible multiclass pipeline
Mota <i>et al.</i> [4]	Q-learning adaptive modulation and coding	CQI state; MCS action	Learns CQI-to-MCS decisions with BLER awareness	Sequential reinforcement learning rather than offline supervised classification
Bobrov <i>et al.</i> [5]	Massive MIMO online deep learning	Service and channel feedback; MCS success or throughput	Demonstrates online adaptation under changing channels	More complex online neural approach, not CQI-only cross-bank XGBoost
Rokoni <i>et al.</i> [6]	Empirical C-V2X sidelink; RF, boosting, and neural models	Drive-test measurements; MCS or data-rate decision	Supports asymmetric treatment of over- and under-prediction	Different sidelink setting and target construction
Balieiro <i>et al.</i> [7]	CQI-delay compensation using ML and evolutionary computing	User context, SINR, and delay; updated SINR/CQI	Shows limitations of stale CQI in dynamic channels	Predicts updated channel quality rather than best MCS
Azari <i>et al.</i> [8]	5G/6G non-terrestrial-network survey	NTN system and channel considerations	Motivates high-Doppler and rain-fade satellite extension	Survey, not an MCS-classification experiment

TABLE II
EVALUATED TERRESTRIAL CONFIGURATION BANKS

Bank	Dataset Name	Configuration Summary
A	BankA_FR1_sub6	FR1 sub-6 GHz configuration
B	BankB_FR2_wide	FR2 wideband configuration
C	BankC_mmW_indoor	Indoor mmWave configuration
D	BankD_mmW_outdoor_LoS	Outdoor line-of-sight mmWave configuration
E	BankE_S_band_FR1	S-band FR1 configuration
F	BankF_L_band_FR1	L-band FR1 configuration
G	BankG_FWA_FR1_suburban	Suburban FR1 fixed wireless access configuration

TABLE III
BANKB FULL-FEATURE MODEL COMPARISON

Model	Exact (%)	± 1 (%)	Over (%)	Under (%)
XGBoost	64.20	88.75	20.03	15.78
DNN	63.08	90.50	16.35	20.58
Decision Tree	61.20	87.98	17.38	21.43
Random Forest	61.00	84.95	20.95	18.05
ExtraTrees	60.85	84.48	20.65	18.50
Linear SVM	56.28	77.85	16.15	27.58
Baseline	46.55	48.28	53.45	0.00

TABLE IV
BANKB TUNED XGBOOST FEATURE COMPARISON

Features	Exact (%)	± 1 (%)	Over (%)	Under (%)
CQI only	65.93	90.63	17.93	16.15
CQI/RI/PMI	65.80	90.38	17.98	16.23
Full	64.20	88.75	20.03	15.78

C. BankB Feature-Set Comparison

CQI-only XGBoost achieves the strongest BankB result. Adding RI and PMI changes performance only marginally, while adding all available channel descriptors slightly reduces exact and within- ± 1 accuracy. This supports the interpretation that CQI already contains sufficient CSI-derived information for this classification task.

D. Cross-Bank CQI-Only XGBoost Evaluation

BankD achieves the highest exact accuracy at 75.88%, while BankG achieves the highest within- ± 1 accuracy at 92.60%.

Both banks also have the lowest combined over-prediction and under-prediction rates. Their high most-frequent-label baselines, approximately 62.7%, indicate that part of the high raw accuracy may be associated with concentrated target distributions.

BankA has the lowest exact accuracy, but it produces the largest gain over baseline at 37.12 percentage points. BankE and BankF also show large gains over baseline. These results show why raw accuracy and improvement over baseline should be interpreted together.

BankD and BankG are both outdoor propagation cases. They use the same TDL-C multipath delay profile and the same SNR range, while differing substantially in carrier frequency, subcarrier spacing, and number of resource blocks. Their similar performance suggests that propagation environment, delay profile, and SNR distribution may be important contributors to predictability, but further controlled analysis is required before isolating the effect of individual parameters.

The substantial variation among FR1 banks further indicates that frequency range alone does not determine classification performance. Propagation conditions, Doppler range, deployment type, subcarrier spacing, number of resource blocks, and target-label distribution may all contribute.

VI. PLANNED SATELLITE EXTENSION

The next stage will extend the CQI-only XGBoost evaluation to LEO satellite communication banks generated under S-band and Ka-band conditions, where mobility, Doppler,

TABLE V
CQI-ONLY TUNED XGBOOST RESULTS ACROSS BANKA–BANKG

Bank	Exact (%)	± 1 (%)	Over (%)	Under (%)	Baseline (%)	Gain (pp)
BankA_FR1_sub6	57.60	87.65	19.82	22.58	20.48	37.12
BankB_FR2_wide	65.93	90.63	17.93	16.15	46.55	19.38
BankC_mmW_indoor	65.80	91.98	19.63	14.58	46.75	19.05
BankD_mmW_outdoor_LoS	75.88	92.58	13.13	11.00	62.78	13.10
BankE_S_band_FR1	64.28	90.20	16.90	18.83	29.55	34.73
BankF_L_band_FR1	62.60	89.05	19.25	18.15	29.35	33.25
BankG_FWA_FR1_suburban	75.05	92.60	14.05	10.90	62.73	12.33

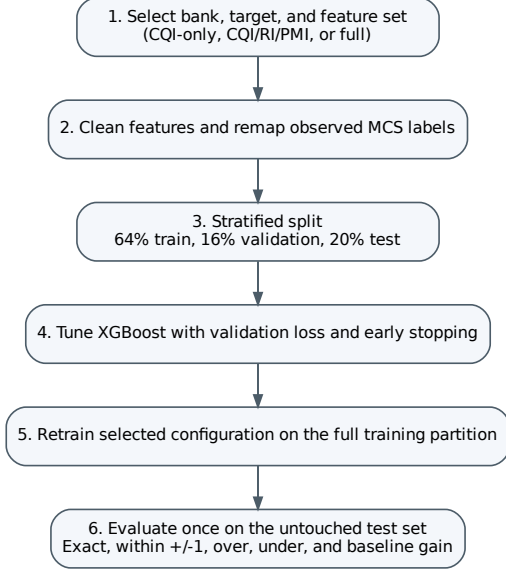


Fig. 2. Validated machine-learning evaluation flow used for configuration selection and final testing.

and time-varying propagation are recognized NTN challenges [8]. Planned cases include clear-sky and rain-fade conditions, 5 MHz and 400 MHz bandwidths, and substantially higher Doppler ranges. The analysis will compare S-band against Ka-band, clear sky against rain fade, and satellite against terrestrial configurations. Particular attention will be given to joint Doppler-induced inter-carrier interference and Ka-band rain-fade impairment.

VII. DISCUSSION

The preliminary experiments show that CQI-only XGBoost is a strong and efficient baseline across all evaluated terrestrial banks. Within- ± 1 accuracy remains above 87% for every bank, while exact accuracy varies more strongly with system configuration and class distribution. This suggests that the model usually predicts an operationally nearby MCS even when the exact class is missed.

The CQI-only result also has a practical interpretation. Since CQI, RI, and PMI are all derived from CSI, RI and PMI may provide redundant rather than independent information for

best-MCS classification. Future analysis should examine per-CQI target distributions, confusion matrices, feature importance, calibration, and bank-specific label imbalance to verify this explanation more rigorously.

VIII. LIMITATIONS AND REPRODUCIBILITY

The current study uses one fixed train/test split and a small manually defined XGBoost configuration set. Before submission, the experiments should be rerun with validation-only hyperparameter selection, multiple random seeds, and mean-plus-standard-deviation reporting. The paper should also include per-bank class distributions, confusion matrices, and the exact configuration metadata used to generate each bank. Claims about the effects of carrier frequency, propagation environment, Doppler, delay profile, subcarrier spacing, and resource-block count should remain associative unless supported by controlled comparisons that vary one factor at a time.

IX. CONCLUSION

A reusable machine-learning pipeline was developed for link adaptation across seven heterogeneous 5G NR system banks. In the preliminary single-split evaluation, tuned XGBoost outperformed the most-frequent-label baseline and achieved strong within- ± 1 MCS accuracy. CQI-only input produced the best BankB result and remained effective across BankA through BankG. The findings indicate that system configuration and target distribution influence exact accuracy, while CQI retains sufficient information for useful best-MCS classification across a broad range of terrestrial scenarios. Future work will extend the analysis to LEO satellite communication datasets under high Doppler and rain-fade conditions.

ACKNOWLEDGMENT

The authors thank the project system team for dataset generation and system-configuration support. Individual contributors will be named in the final version after their full names and preferred attribution are confirmed.

REFERENCES

- [1] L. Tsipi, M. Karavolos, G. Papaioannou, M. Volakaki, and D. Vouyioukas, "Machine learning-based methods for MCS prediction in 5G networks," *Telecommunication Systems*, vol. 86, pp. 705–728, 2024, doi: 10.1007/s11235-024-01158-x.
- [2] P. S. Pati, S. S. Sahoo, D. Krishnaswamy, and R. Datta, "A novel machine learning approach for link adaptation in 5G wireless networks," in *Proc. 2nd PhD Colloquium on Ethically Driven Innovation and Technology for Society (PhD EDITS)*, 2020, pp. 1–2, doi: 10.1109/PhDED-ITS51180.2020.9315299.

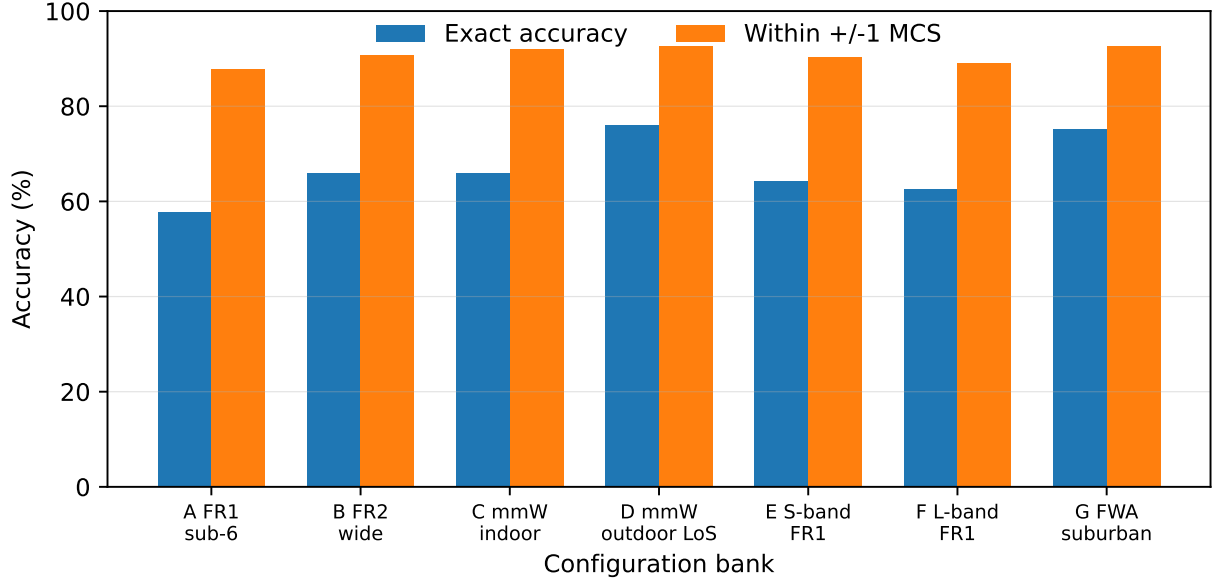


Fig. 3. Preliminary CQI-only tuned-XGBoost exact and within- ± 1 accuracy across BankA–BankG. These values should be updated after the validation-only rerun described in Section IV-C.

- [3] F. J. Martín-Vega, J. C. Ruiz-Sicilia, M. C. Aguayo-Torres, and G. Gómez, “Emerging tools for link adaptation on 5G NR and beyond: Challenges and opportunities,” *IEEE Access*, vol. 9, pp. 126976–126987, 2021, doi: 10.1109/ACCESS.2021.3111783.
- [4] M. P. Mota, D. C. Araujo, F. H. Costa Neto, A. L. F. de Almeida, and F. R. P. Cavalcanti, “Adaptive modulation and coding based on reinforcement learning for 5G networks,” in *Proc. IEEE Globecom Workshops*, 2019, pp. 1–6, doi: 10.1109/GCWkshps45667.2019.9024384.
- [5] E. Bobrov, D. Kropotov, H. Lu, and D. Zaev, “Massive MIMO adaptive modulation and coding using online deep learning algorithm,” *IEEE Communications Letters*, vol. 26, no. 4, pp. 818–822, Apr. 2022, doi: 10.1109/LCOMM.2021.3132947.
- [6] A. A. Rokoni, D. Schäufele, M. Kasparick, and S. Stańczak, “From empirical measurements to augmented data rates: A machine learning approach for MCS adaptation in sidelink communication,” in *Proc. IEEE 98th Vehicular Technology Conference (VTC2023-Fall)*, 2023, pp. 1–5, doi: 10.1109/VTC2023-Fall60731.2023.10333593.
- [7] A. Balieiro, K. Dias, and P. Guarda, “Addressing the CQI feedback delay in 5G/6G networks via machine learning and evolutionary computing,” *Intelligent and Converged Networks*, vol. 3, no. 3, pp. 271–281, 2022, doi: 10.23919/ICN.2022.0012.
- [8] M. M. Azari *et al.*, “Evolution of non-terrestrial networks from 5G to 6G: A survey,” *IEEE Communications Surveys & Tutorials*, vol. 24, no. 4, pp. 2633–2672, 2022, doi: 10.1109/COMST.2022.3199901.
- [9] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.
- [10] 3GPP, “NR; physical layer procedures for data,” 3GPP TS 38.214, Release 18. [Online]. Available: <https://www.3gpp.org/dynareport/38214.htm>